

Finite Element Methods – APE2

1D FEM - **Solutions**

1 Can a finite element solution be nodally exact ?

Consider the Poisson problem

$$-\frac{d}{dx}\left(\kappa(x)\frac{du}{dx}\right) = f(x) \quad \text{in } \Omega = (0, 1), \quad (1)$$

with the boundary conditions

$$u(0) = 0, \quad u(1) = 0$$

1. Check with the code that the the finite element solution is nodally exact in the case $\kappa(x) = 1$ and $f(x) = 1$. Is it still exact for $(\kappa(x) = 1+x, f(x) = 1)$, $(\kappa(x) = 1, f(x) = 1+x)$, $(\kappa(x) = 1+x, f(x) = 1+x)$?
2. Prove that 1D finite elements do give nodally exact solution when κ does not depend on x .

Solutions

Verifying if the numerical solution is nodally exact or not requires the analytical solution. We will consider this general case : $\kappa(x) = 1 + ax$ and $f(x) = 1 + bx$, $a \geq 0, b \geq 0$.

$$-((1+ax)u')' = 1+bx \quad \rightarrow \quad -(1+ax)u' = x + \frac{b}{2}x^2 + C_1 \quad \rightarrow \quad u' = -\frac{x + \frac{b}{2}x^2 + C_1}{1+ax}$$

If $a \neq 0$

$$u(x) = -\int \frac{x + \frac{b}{2}x^2 + C_1}{1+ax} dx = -\frac{b}{4a}x^2 - \left(\frac{1}{a} - \frac{b}{2a^2}\right)x - \frac{1}{a}\left(C_1 - \frac{1}{a} + \frac{b}{2a^2}\right)\ln|1+ax| + C_2$$

$$C_1 = \frac{1}{a} - \frac{b}{2a^2} - \frac{4a+ba-2b}{4a \ln|1+a|}, \quad C_2 = 0$$

$$u(x) = -\frac{b}{4a}x^2 - \left(\frac{1}{a} - \frac{b}{2a^2}\right)x + \frac{1}{a}\left(\frac{4a+ba-2b}{4a \ln|1+a|}\right)\ln|1+ax|$$

If $a = 0$

$$u(x) = -\int x + \frac{b}{2}x^2 + C_1 dx = -\frac{x^2}{2} - \frac{b}{6}x^3 + C_1x + C_2$$

$$C_1 = \frac{1}{2} + \frac{b}{6}, \quad C_2 = 0$$

So,

$$u(x) = -\frac{x^2}{2} - \frac{b}{6}x^3 + \left(\frac{1}{2} + \frac{b}{6}\right)x$$

In particular:

(a, b)	$u(x)$
$(0, 0)$	$-\frac{x^2}{2} + \frac{x}{2} = \frac{x(1-x)}{2}$
$(0, 1)$	$-\frac{x^2}{2} - \frac{x^3}{6} + \frac{2}{3}x = -\frac{x(x-1)(x+4)}{6}$
$(1, 0)$	$-x + \frac{1}{\ln 2} \ln(1+x)$
$(1, 1)$	$-\frac{x^2}{4} - \frac{x}{2} + \frac{3}{4\ln 2} \ln(1+x)$

The finite element solution should coincide with the exact solution at the nodes only in the first two cases, **provided that** the right-hand side is integrated exactly. Since we are integrating a quadratic function, two Gauss points are required. If there is only one Gauss point, the solution should not coincide.

Regarding the proof, we write that the exact solution satisfies the variational principle for all test functions in H^1 , and therefore also for the finite element subspace of test functions. We also state that the finite element solution satisfies the variational principle for the finite element test functions.

$$\begin{aligned} \int_0^1 \kappa(x) u' v' - f(x) v dx &= 0, \quad \forall v \in U_h^0 \\ \int_0^1 \kappa(x) u_h' v' - f(x) v dx &= 0, \quad \forall v \in U_h^0 \end{aligned}$$

So, taking the difference, we get

$$\int_0^1 \kappa(x) (u - u_h)' v' dx = 0, \quad \forall v \in U_h^0$$

The solution u_h is unique (because it is a linear problem). We check if $u_h = I_h u$ is solution where $I_h u$ is the nodally exact interpolation. Assuming $\kappa(x)$ uniform, we are going to show that the above integral is zero over each element. We have using integration by parts

$$\begin{aligned} \int_{x_i}^{x_{i+1}} (u - I_h u)' (\kappa(x) v') dx &= - \int_{x_i}^{x_{i+1}} (u - I_h u) (\kappa(x) v')' dx + [(u - I_h u) (\kappa(x) v')]_i^{i+1} \\ &= - \int_{x_i}^{x_{i+1}} (u - I_h u) (\kappa' v') dx + 0 \\ &= 0 \end{aligned}$$

Remark: Students should understand that $U_h \subset U$, or $U_h^0 \subset U^0$. And, since Dirichlet conditions are zero, U^0 and U are the same. Students should also understand why in higher dimension, nodally exact FEM solution is not very likely to happen.

2 Finite element convergence

We continue with the choice $\kappa(x) = 1, f(x) = 1$.

1. Compute the exact energy
2. With the code, get the finite element energy.
3. Check that finite element energy is always above the exact one.
4. Analyze the way the energy is converging with the mesh size.

Solutions

The exact solution is

$$u(x) = \frac{1}{2}x(1-x), u'(x) = 1/2 - x$$

The exact energy is

$$\Pi(u) = \int_0^1 \frac{1}{2}(1/2 - x)^2 - \frac{1}{2}x(1-x)dx = -1/24$$

Students need to check that for any mesh, the finite element energy is above the exact one whatever the order of the finite elements (but you can just stay with linear elements). In the classroom, we have seen that :

$$\Pi(u_h) - \Pi(u) = \frac{1}{2}\|u_h - u\|_a^2 \quad (2)$$

Where the a subscript indicates the energy norm defined by

$$\|u\|_a^2 = a(u, u), \quad a(u, v) = \int_0^L \kappa(x)u'v'dx$$

This energy norm is different from the H^1 seminorm given by

$$\|u\|_{H^{1s}(0,1)}^2 = \int_0^L (u')^2 dx$$

Note that both norms are identical when $\kappa(x) = 1$ which is the case in this exercise. Note that the error is simply the (square root of) the difference in energy. This gives a convenient way to obtain the error associated to a finite element solution.

The bilinear operator $a(\cdot, \cdot)$ and linear operator $\ell(\cdot)$ appear in the weak formulation and the energy functional:

$$\begin{aligned} \Pi(u) &= \frac{1}{2}a(u, u) - \ell(u), \quad \ell(u) = \int_0^L f u dx \\ a(u, v) &= \ell(v), \forall v \in V \end{aligned}$$

We can prove 2, by developing the energy gap.

$$\begin{aligned} \Pi(u_h) - \Pi(u) &= \frac{1}{2}a(u_h, u_h) - \ell(u_h) - \frac{1}{2}a(u, u) + \ell(u) \\ &= \frac{1}{2}(a(u_h, u_h) - a(u, u)) - \ell(u_h - u) \end{aligned}$$

As $u_h - u \in V$, we have $a(u, u_h - u) = \ell(u_h - u)$. And due to bilinearity, we can write $a(u, u_h - u) = a(u, u_h) - a(u, u)$. By plugging it back, we obtain :

$$\begin{aligned} \Pi(u_h) - \Pi(u) &= \frac{1}{2}(a(u_h, u_h) - a(u, u)) - a(u, u_h) + a(u, u) \\ &= \frac{1}{2}(a(u_h, u_h) - 2a(u, u_h) + a(u, u)) \\ &= \frac{1}{2}a(u_h - u, u_h - u) = \frac{1}{2}\|u_h - u\|_a^2 \end{aligned}$$

Students can then consider several converging sequence of meshes. Each sequence is obtained by dividing by 2 at each step, the left and right imposed mesh size. For each sequence, the error

will monotonically go down since each finite element space is larger than the previous one. Students should observe that uniform mesh sequence is the one with the smallest error. (For the problem at stake, the uniform mesh gives for a given number of nodes the least energy).

Make sure that the student draw the error in a log-log scale to see the order of convergence (which should be one for linear elements) and the asymptotic constant. More precisely,

$$\|u_h - u\|_a = O(h) \text{ meaning } = Ch \text{ as } h \rightarrow 0^+ \rightarrow \log(\|u_h - u\|_a) = \log h + \log C$$

where C is a constant independent of the mesh size but depending on the mesh sequence (uniform or not). For all mesh sequence, the log-log plot should all be roughly linear (and more and more so as h gets smaller) and the lowest line is for the uniform mesh.

Bonus : You can also have students find the middle node location minimizing the energy for just a two elements bar problem. They should find $x = 0.5$.

3 A more complex energy

Consider the following energy

$$\Pi(u) = \int_{r_0}^{r_1} r (u')^2 + \frac{1}{4} \frac{u^2}{r} dr \quad (3)$$

with Dirichlet conditions

$$u(r_0) = u_0, \quad u(r_1 = 1) = 1$$

where $0 \leq r_0 \leq 1$.

1. Find the associate finite element problem.
2. Modify the code so that it can analyze this new energy function.
3. Compare the finite element solution to the exact one, and the finite element energy to exact one.
4. What happens if $r_0 = 0$ and how can you handle it in the finite element formulation?

Solutions

The finite element matrix is given by

$$K_{ij} = \int_0^1 2r \varphi'_i \varphi'_j + \frac{1}{2} \frac{\varphi_i \varphi_j}{r} dr$$

in which Dirichlet boundary conditions need to be taken into account. It is thus the sum of a stiffness matrix and a mass matrix with special weights.

To find the exact solution, we have

$$-(2ru')' + \frac{1}{2} \frac{u}{r} = 0$$

By choosing $u = r^m$, we find $m = 1/2$ or $-1/2$. So

$$u(r) = \frac{C_0}{\sqrt{r}} + C_1 \sqrt{r}$$

Applying the boundary conditions, gives

$$u(r) = \frac{(1 - u_0\sqrt{r_0})\sqrt{r} + (u_0\sqrt{r_0} - r_0)r^{-1/2}}{1 - r_0}$$

As r_0 goes to zero, the solution goes to $\sqrt{r}$ and the only boundary conditions that we can impose at $r = 0$ is zero to ensure a finite energy.

When $r_0 = 0$, the mass matrix is integrable because there is no degree of freedom associated to the node at $r = 0$.